MAT120: Integral Calculus

Exercise Set 7.3 — Trigonometric Integrals (Problems 23–50)

Step-by-Step Detailed Solutions Grounded in Trigonometric Identities & Reference Rules

Formula Reference Guide (from formula.png)

Integration Rules for Products of Tangents and Secants $\int \tan^m x \sec^n x\,dx$:

1. **If n is even:** Save a factor of $\sec^2 x$, convert the remaining secants using $\sec^2 x = \tan^2 x + 1$, and substitute $u = \tan x$ ($du = \sec^2 x\,dx$).
2. **If m is odd:** Save a factor of $\sec x \tan x$, convert the remaining tangents using $\tan^2 x = \sec^2 x - 1$, and substitute $u = \sec x$ ($du = \sec x \tan x\,dx$).
3. **If m is even and n is odd:** Convert all tangents to secants using $\tan^2 x = \sec^2 x - 1$, then integrate powers of secant using reduction formulas or integration by parts.

Standard Integrals Reference:

$$\int \tan x\,dx = \ln|\sec x| + C, \qquad \int \sec x\,dx = \ln|\sec x + \tan x| + C$$

$$\int \sec^3 x\,dx = \frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x| + C$$

$$\int \sec^5 x\,dx = \frac{1}{4}\sec^3 x \tan x + \frac{3}{8}\sec x \tan x + \frac{3}{8}\ln|\sec x + \tan x| + C$$

Part I: Indefinite Integrals (Problems 23–44, 49–50)

Problem 23: Evaluate $\int \sec^2(2x-1)\,dx$

Method / Strategy: Direct substitution using standard integral $\int \sec^2 u\,du = \tan u + C$.
Step 1: Let $u = 2x - 1$. Then $du = 2\,dx \implies dx = \frac{1}{2}du$.
Step 2: Substitute into the integral:

$$\int \sec^2(2x-1)\,dx = \int \sec^2 u \left(\frac{1}{2}du\right) = \frac{1}{2}\int \sec^2 u\,du = \frac{1}{2}\tan u + C$$

Step 3: Substitute back $u = 2x - 1$:

$$\boxed{\int \sec^2(2x-1)\,dx = \frac{1}{2}\tan(2x-1) + C}$$

Problem 24: Evaluate $\int \tan 6x\,dx$

Method / Strategy: Rewrite $\tan 6x = \frac{\sin 6x}{\cos 6x}$ and use substitution $u = \cos 6x$.
Step 1: Let $u = \cos 6x$. Then $du = -6\sin 6x\,dx \implies \sin 6x\,dx = -\frac{1}{6}du$.
Step 2: Integrate:

$$\int \tan 6x\,dx = \int \frac{\sin 6x}{\cos 6x}\,dx = \int \frac{-\frac{1}{6}du}{u} = -\frac{1}{6}\ln|u| + C = \frac{1}{6}\ln|\sec 6x| + C$$

Step 3: Final Boxed Answer:

$$\boxed{\int \tan 6x\,dx = \frac{1}{6}\ln|\sec 6x| + C}$$

Problem 25: Evaluate $\int e^{-x}\tan(e^{-x})\,dx$

Method / Strategy: Substitution $u = e^{-x}$.
Step 1: Let $u = e^{-x}$. Then $du = -e^{-x}\,dx \implies e^{-x}\,dx = -du$.
Step 2: Substitute and integrate:

$$\int e^{-x}\tan(e^{-x})\,dx = \int \tan u(-du) = -\int \tan u\,du = -\ln|\sec u| + C = \ln|\cos u| + C$$

Step 3: Substitute back $u = e^{-x}$:

$$\boxed{\int e^{-x}\tan(e^{-x})\,dx = \ln\left|\cos(e^{-x})\right| + C \quad \text{or} \quad -\ln\left|\sec(e^{-x})\right| + C}$$

Problem 26: Evaluate $\int \cot 4x\,dx$

Method / Strategy: Rewrite $\cot 4x = \frac{\cos 4x}{\sin 4x}$ and use substitution $u = \sin 4x$.
Step 1: Let $u = \sin 4x$. Then $du = 4\cos 4x\,dx \implies \cos 4x\,dx = \frac{1}{4}du$.
Step 2: Integrate:

$$\int \cot 4x\,dx = \int \frac{\frac{1}{4}du}{u} = \frac{1}{4}\ln|u| + C = \frac{1}{4}\ln|\sin 4x| + C$$

Step 3: Final Boxed Answer:

$$\boxed{\int \cot 4x\,dx = \frac{1}{4}\ln|\sin 4x| + C}$$

Problem 27: Evaluate $\int \sec 7x\,dx$

Method / Strategy: Substitution $u = 7x$ with standard formula $\int \sec u\,du = \ln|\sec u + \tan u| + C$.
Step 1: Let $u = 7x$. Then $du = 7\,dx \implies dx = \frac{1}{7}du$.
Step 2: Integrate:

$$\int \sec 7x\,dx = \frac{1}{7}\int \sec u\,du = \frac{1}{7}\ln|\sec u + \tan u| + C$$

Step 3: Substitute back $u = 7x$:

$$\boxed{\int \sec 7x\,dx = \frac{1}{7}\ln|\sec 7x + \tan 7x| + C}$$

Problem 28: Evaluate $\displaystyle\int \frac{\sec(\sqrt{x})}{\sqrt{x}}\,dx$

Method / Strategy: Substitution $u = \sqrt{x}$.

Step 1: Let $u = \sqrt{x}$. Then $du = \frac{1}{2\sqrt{x}}\,dx \implies \frac{dx}{\sqrt{x}} = 2du$.

Step 2: Substitute into integral:

$$\int \frac{\sec(\sqrt{x})}{\sqrt{x}}\,dx = \int \sec u(2du) = 2\int \sec u\,du = 2\ln|\sec u + \tan u| + C$$

Step 3: Substitute back $u = \sqrt{x}$:

$$\boxed{\int \frac{\sec(\sqrt{x})}{\sqrt{x}}\,dx = 2\ln\left|\sec(\sqrt{x}) + \tan(\sqrt{x})\right| + C}$$

Problem 29: Evaluate $\displaystyle\int \tan^2 x \sec^2 x\,dx$

Method / Strategy: `formula.png` Case 1 ($n = 2$ is even). Substitute $u = \tan x$.

Step 1: Let $u = \tan x$. Then $du = \sec^2 x\,dx$.

Step 2: Substitute and integrate:

$$\int \tan^2 x \sec^2 x\,dx = \int u^2\,du = \frac{u^3}{3} + C$$

Step 3: Substitute back $u = \tan x$:

$$\boxed{\int \tan^2 x \sec^2 x\,dx = \frac{1}{3}\tan^3 x + C}$$

Problem 30: Evaluate $\displaystyle\int \tan^3 x \sec^4 x\,dx$

Method / Strategy: `formula.png` Case 1 ($n = 4$ is even) or Case 2 ($m = 3$ is odd). Using Case 1 with $u = \tan x$.

Step 1: Split $\sec^4 x = \sec^2 x \cdot \sec^2 x = (\tan^2 x + 1)\sec^2 x$.

Step 2: Let $u = \tan x \implies du = \sec^2 x\,dx$:

$$\int \tan^3 x(\tan^2 x + 1)\sec^2 x\,dx = \int u^3(u^2+1)\,du = \int (u^5 + u^3)\,du = \frac{u^6}{6} + \frac{u^4}{4} + C$$

Step 3: Substitute back $u = \tan x$:

$$\boxed{\int \tan^3 x \sec^4 x\,dx = \frac{1}{6}\tan^6 x + \frac{1}{4}\tan^4 x + C}$$

Problem 31: Evaluate $\int \tan 4x \sec^4 4x\,dx$

Method / Strategy: `formula.png` Case 2 ($m = 1$ is odd). Substitute $u = \sec 4x$.
Step 1: Rewrite integrand as $\sec^3 4x \cdot (\sec 4x \tan 4x)$.
Step 2: Let $u = \sec 4x \implies du = 4\sec 4x \tan 4x\,dx \implies \sec 4x \tan 4x\,dx = \frac{1}{4}du$.
Step 3: Substitute and integrate:

$$\int u^3 \left(\frac{1}{4}du\right) = \frac{1}{4}\cdot\frac{u^4}{4} + C = \frac{1}{16}u^4 + C = \frac{1}{16}\sec^4 4x + C$$

$$\boxed{\int \tan 4x \sec^4 4x\,dx = \frac{1}{16}\sec^4 4x + C}$$

Problem 32: Evaluate $\int \tan^4\theta \sec^4\theta\,d\theta$

Method / Strategy: `formula.png` Case 1 ($n = 4$ is even). Substitute $u = \tan\theta$.
Step 1: Rewrite $\sec^4\theta = (\tan^2\theta + 1)\sec^2\theta$.
Step 2: Let $u = \tan\theta \implies du = \sec^2\theta\,d\theta$:

$$\int \tan^4\theta(\tan^2\theta + 1)\sec^2\theta\,d\theta = \int u^4(u^2+1)\,du = \int (u^6 + u^4)\,du = \frac{u^7}{7} + \frac{u^5}{5} + C$$

Step 3: Substitute back $u = \tan\theta$:

$$\boxed{\int \tan^4\theta \sec^4\theta\,d\theta = \frac{1}{7}\tan^7\theta + \frac{1}{5}\tan^5\theta + C}$$

Problem 33: Evaluate $\int \sec^3 x \tan^3 x\,dx$

Method / Strategy: `formula.png` Case 2 ($m = 3$ is odd). Substitute $u = \sec x$.
Step 1: Save factor $\sec x \tan x$ and convert $\tan^2 x = \sec^2 x - 1$:

$$\sec^3 x \tan^3 x = \sec^2 x(\sec^2 x - 1)(\sec x \tan x)$$

Step 2: Let $u = \sec x \implies du = \sec x \tan x\,dx$:

$$\int u^2(u^2 - 1)\,du = \int (u^4 - u^2)\,du = \frac{u^5}{5} - \frac{u^3}{3} + C$$

Step 3: Substitute back $u = \sec x$:

$$\boxed{\int \sec^3 x \tan^3 x\,dx = \frac{1}{5}\sec^5 x - \frac{1}{3}\sec^3 x + C}$$

Problem 34: Evaluate $\int \tan^5\theta \sec\theta \, d\theta$

Method / Strategy: `formula.png` Case 2 ($m = 5$ is odd). Substitute $u = \sec\theta$.
Step 1: Save factor $\sec\theta\tan\theta$ and convert $\tan^4\theta = (\sec^2\theta - 1)^2$:

$$\tan^5\theta\sec\theta = (\sec^2\theta - 1)^2(\sec\theta\tan\theta)$$

Step 2: Let $u = \sec\theta \implies du = \sec\theta\tan\theta\, d\theta$:

$$\int (u^2-1)^2\, du = \int (u^4 - 2u^2 + 1)\, du = \frac{u^5}{5} - \frac{2u^3}{3} + u + C$$

Step 3: Substitute back $u = \sec\theta$:

$$\boxed{\int \tan^5\theta\sec\theta\, d\theta = \frac{1}{5}\sec^5\theta - \frac{2}{3}\sec^3\theta + \sec\theta + C}$$

Problem 35: Evaluate $\int \tan^4 x \sec x \, dx$

Method / Strategy: `formula.png` Case 3 ($m = 4$ is even, $n = 1$ is odd). Convert $\tan^4 x = (\sec^2 x - 1)^2$.
Step 1: Expand integrand:

$$\tan^4 x\sec x = (\sec^2 x - 1)^2\sec x = \sec^5 x - 2\sec^3 x + \sec x$$

Step 2: Integrate term-by-term using reference formulas:

$$\int \tan^4 x\sec x\, dx = \int \sec^5 x\, dx - 2\int \sec^3 x\, dx + \int \sec x\, dx$$

Step 3: Substitute values and simplify coefficients:

$$\begin{aligned}
&= \left(\frac{1}{4}\sec^3 x\tan x + \frac{3}{8}\sec x\tan x + \frac{3}{8}\ln|\sec x + \tan x|\right) \\
&\quad - 2\left(\frac{1}{2}\sec x\tan x + \frac{1}{2}\ln|\sec x + \tan x|\right) + \ln|\sec x + \tan x| + C \\
&= \frac{1}{4}\sec^3 x\tan x - \frac{5}{8}\sec x\tan x + \frac{3}{8}\ln|\sec x + \tan x| + C
\end{aligned}$$

$$\boxed{\int \tan^4 x\sec x\, dx = \frac{1}{4}\sec^3 x\tan x - \frac{5}{8}\sec x\tan x + \frac{3}{8}\ln|\sec x + \tan x| + C}$$

Problem 36: Evaluate $\int \tan^2 x \sec^3 x \, dx$

Method / Strategy: `formula.png` Case 3 ($m = 2$ even, $n = 3$ odd). Convert $\tan^2 x = \sec^2 x - 1$.

Step 1: Rewrite integrand:

$$\tan^2 x \sec^3 x = (\sec^2 x - 1)\sec^3 x = \sec^5 x - \sec^3 x$$

Step 2: Use formulas for $\int \sec^5 x \, dx$ and $\int \sec^3 x \, dx$:

$$\int \tan^2 x \sec^3 x \, dx = \left(\frac{1}{4}\sec^3 x \tan x + \frac{3}{8}\sec x \tan x + \frac{3}{8}\ln|\sec x + \tan x|\right) - \left(\frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x|\right) + C$$

Step 3: Simplify coefficients:

$$\boxed{\int \tan^2 x \sec^3 x \, dx = \frac{1}{4}\sec^3 x \tan x - \frac{1}{8}\sec x \tan x - \frac{1}{8}\ln|\sec x + \tan x| + C}$$

Problem 37: Evaluate $\int \tan t \sec^3 t \, dt$

Method / Strategy: `formula.png` Case 2 ($m = 1$ is odd). Substitute $u = \sec t$.

Step 1: Rewrite as $\sec^2 t(\sec t \tan t)$.

Step 2: Let $u = \sec t \implies du = \sec t \tan t \, dt$:

$$\int u^2 \, du = \frac{u^3}{3} + C = \frac{1}{3}\sec^3 t + C$$

Step 3: Final Boxed Answer:

$$\boxed{\int \tan t \sec^3 t \, dt = \frac{1}{3}\sec^3 t + C}$$

Problem 38: Evaluate $\int \tan x \sec^5 x \, dx$

Method / Strategy: `formula.png` Case 2 ($m = 1$ is odd). Substitute $u = \sec x$.

Step 1: Rewrite as $\sec^4 x(\sec x \tan x)$.

Step 2: Let $u = \sec x \implies du = \sec x \tan x \, dx$:

$$\int u^4 \, du = \frac{u^5}{5} + C = \frac{1}{5}\sec^5 x + C$$

Step 3: Final Boxed Answer:

$$\boxed{\int \tan x \sec^5 x \, dx = \frac{1}{5}\sec^5 x + C}$$

Problem 39: Evaluate $\int \sec^4 x\,dx$

Method / Strategy: `formula.png` Case 1 ($n = 4$ is even). Substitute $u = \tan x$.

Step 1: Rewrite $\sec^4 x = (\tan^2 x + 1)\sec^2 x$.

Step 2: Let $u = \tan x \implies du = \sec^2 x\,dx$:

$$\int (u^2 + 1)\,du = \frac{u^3}{3} + u + C = \frac{1}{3}\tan^3 x + \tan x + C$$

Step 3: Final Boxed Answer:

$$\boxed{\int \sec^4 x\,dx = \frac{1}{3}\tan^3 x + \tan x + C}$$

Problem 40: Evaluate $\int \sec^5 x\,dx$

Method / Strategy: Integration by parts or reduction formula $\int \sec^n x\,dx = \frac{1}{n-1}\sec^{n-2} x\tan x + \frac{n-2}{n-1}\int \sec^{n-2} x\,dx$.

Step 1: Apply reduction formula with $n = 5$:

$$\int \sec^5 x\,dx = \frac{1}{4}\sec^3 x\tan x + \frac{3}{4}\int \sec^3 x\,dx$$

Step 2: Substitute $\int \sec^3 x\,dx = \frac{1}{2}\sec x\tan x + \frac{1}{2}\ln|\sec x + \tan x|$:

$$\int \sec^5 x\,dx = \frac{1}{4}\sec^3 x\tan x + \frac{3}{4}\left(\frac{1}{2}\sec x\tan x + \frac{1}{2}\ln|\sec x + \tan x|\right) + C$$

Step 3: Simplify terms:

$$\boxed{\int \sec^5 x\,dx = \frac{1}{4}\sec^3 x\tan x + \frac{3}{8}\sec x\tan x + \frac{3}{8}\ln|\sec x + \tan x| + C}$$

Problem 41: Evaluate $\int \tan^3 4x\,dx$

Method / Strategy: Separate $\tan^2 4x = \sec^2 4x - 1$.

Step 1: Rewrite integrand:

$$\tan^3 4x = \tan 4x(\sec^2 4x - 1) = \tan 4x\sec^2 4x - \tan 4x$$

Step 2: Integrate term-by-term:

$$\int \tan 4x\sec^2 4x\,dx = \frac{1}{8}\tan^2 4x \quad (\text{let } u = \tan 4x)$$

$$\int \tan 4x\,dx = \frac{1}{4}\ln|\sec 4x|$$

Step 3: Combine:

$$\boxed{\int \tan^3 4x\,dx = \frac{1}{8}\tan^2 4x - \frac{1}{4}\ln|\sec 4x| + C}$$

Problem 42: Evaluate $\int \tan^4 x\,dx$

Method / Strategy: Convert $\tan^2 x = \sec^2 x - 1$.

Step 1: Rewrite integrand:

$$\tan^4 x = \tan^2 x(\sec^2 x - 1) = \tan^2 x \sec^2 x - \tan^2 x = \tan^2 x \sec^2 x - (\sec^2 x - 1)$$

Step 2: Integrate term-by-term:

$$\int \tan^2 x \sec^2 x\,dx = \frac{1}{3}\tan^3 x$$

$$\int (\sec^2 x - 1)\,dx = \tan x - x$$

Step 3: Subtract components:

$$\boxed{\int \tan^4 x\,dx = \frac{1}{3}\tan^3 x - \tan x + x + C}$$

Problem 43: Evaluate $\int \sqrt{\tan x}\sec^4 x\,dx$

Method / Strategy: `formula.png` Case 1 ($n = 4$ is even). Substitute $u = \tan x$.

Step 1: Rewrite $\sec^4 x = (\tan^2 x + 1)\sec^2 x$.

Step 2: Let $u = \tan x \implies du = \sec^2 x\,dx$:

$$\int u^{1/2}(u^2 + 1)\,du = \int (u^{5/2} + u^{1/2})\,du = \frac{u^{7/2}}{7/2} + \frac{u^{3/2}}{3/2} + C = \frac{2}{7}u^{7/2} + \frac{2}{3}u^{3/2} + C$$

Step 3: Substitute back $u = \tan x$:

$$\boxed{\int \sqrt{\tan x}\sec^4 x\,dx = \frac{2}{7}\tan^{7/2} x + \frac{2}{3}\tan^{3/2} x + C}$$

Problem 44: Evaluate $\int \tan x \sec^{3/2} x\,dx$

Method / Strategy: `formula.png` Case 2 ($m = 1$ is odd). Substitute $u = \sec x$.

Step 1: Rewrite integrand as $\sec^{1/2} x(\sec x \tan x)$.

Step 2: Let $u = \sec x \implies du = \sec x \tan x\,dx$:

$$\int u^{1/2}\,du = \frac{u^{3/2}}{3/2} + C = \frac{2}{3}u^{3/2} + C = \frac{2}{3}\sec^{3/2} x + C$$

Step 3: Final Boxed Answer:

$$\boxed{\int \tan x \sec^{3/2} x\,dx = \frac{2}{3}\sec^{3/2} x + C}$$

Problem 49: Evaluate $\int \cot^3 x \csc^3 x \, dx$

Method / Strategy: Analogous to tangent-secant Case 2 ($m = 3$ is odd). Substitute $u = \csc x$.

Step 1: Save factor $\csc x \cot x$ and convert $\cot^2 x = \csc^2 x - 1$:

$$\cot^3 x \csc^3 x = \csc^2 x(\csc^2 x - 1)(\csc x \cot x)$$

Step 2: Let $u = \csc x \implies du = -\csc x \cot x \, dx \implies \csc x \cot x \, dx = -du$:

$$\int u^2(u^2 - 1)(-du) = \int (u^2 - u^4)\, du = \frac{u^3}{3} - \frac{u^5}{5} + C$$

Step 3: Substitute back $u = \csc x$:

$$\boxed{\int \cot^3 x \csc^3 x \, dx = \frac{1}{3}\csc^3 x - \frac{1}{5}\csc^5 x + C}$$

Problem 50: Evaluate $\int \cot^2 3t \sec 3t \, dt$

Method / Strategy: Express in terms of sine and cosine.

Step 1: Simplify the integrand:

$$\cot^2 3t \sec 3t = \frac{\cos^2 3t}{\sin^2 3t} \cdot \frac{1}{\cos 3t} = \frac{\cos 3t}{\sin^2 3t} = \cot 3t \csc 3t$$

Step 2: Substitute $u = \sin 3t \implies du = 3\cos 3t \, dt \implies \cos 3t \, dt = \frac{1}{3}du$:

$$\int \frac{\frac{1}{3}du}{u^2} = \frac{1}{3}\left(-\frac{1}{u}\right) + C = -\frac{1}{3\sin 3t} + C = -\frac{1}{3}\csc 3t + C$$

Step 3: Final Boxed Answer:

$$\boxed{\int \cot^2 3t \sec 3t \, dt = -\frac{1}{3}\csc 3t + C}$$

Part II: Definite Integrals (Problems 45–48)

Problem 45: Evaluate $\displaystyle\int_0^{\pi/8} \tan^2 2x\,dx$

Method / Strategy: Use identity $\tan^2 2x = \sec^2 2x - 1$.

Step 1: Rewrite the integral:

$$\int_0^{\pi/8} (\sec^2 2x - 1)\,dx = \left[\frac{1}{2}\tan 2x - x\right]_0^{\pi/8}$$

Step 2: Evaluate at upper limit $x = \pi/8$:

$$\frac{1}{2}\tan\left(2\cdot\frac{\pi}{8}\right) - \frac{\pi}{8} = \frac{1}{2}\tan\left(\frac{\pi}{4}\right) - \frac{\pi}{8} = \frac{1}{2}(1) - \frac{\pi}{8} = \frac{1}{2} - \frac{\pi}{8}$$

Step 3: Evaluate at lower limit $x = 0$:

$$\frac{1}{2}\tan(0) - 0 = 0$$

Step 4: Final Boxed Answer:

$$\boxed{\int_0^{\pi/8} \tan^2 2x\,dx = \frac{1}{2} - \frac{\pi}{8}}$$

Problem 46: Evaluate $\displaystyle\int_0^{\pi/6} \sec^3 2\theta \tan 2\theta\,d\theta$

Method / Strategy: `formula.png` Case 2 ($m = 1$ is odd). Substitute $u = \sec 2\theta$.

Step 1: Rewrite integrand as $\sec^2 2\theta(\sec 2\theta \tan 2\theta)$.

Step 2: Let $u = \sec 2\theta \implies du = 2\sec 2\theta \tan 2\theta\,d\theta \implies \sec 2\theta \tan 2\theta\,d\theta = \frac{1}{2}du$.

Step 3: Change limits of integration:

- Lower limit: $\theta = 0 \implies u = \sec(0) = 1$
- Upper limit: $\theta = \pi/6 \implies u = \sec(2\cdot\pi/6) = \sec(\pi/3) = 2$

Step 4: Evaluate definite integral:

$$\int_1^2 u^2\left(\frac{1}{2}du\right) = \frac{1}{2}\left[\frac{u^3}{3}\right]_1^2 = \frac{1}{6}(2^3 - 1^3) = \frac{1}{6}(8-1) = \frac{7}{6}$$

$$\boxed{\int_0^{\pi/6} \sec^3 2\theta \tan 2\theta\,d\theta = \frac{7}{6}}$$

Problem 47: Evaluate $\displaystyle\int_0^{\pi/2} \tan^5 \frac{x}{2}\,dx$

Method / Strategy: `formula.png` Case 2 ($m = 5$ is odd) or reduction. Substitute $w = x/2 \implies dx = 2dw$.

Step 1: Transform integral with $w = x/2$: limits $x = 0 \to w = 0$, $x = \pi/2 \to w = \pi/4$:

$$I = 2\int_0^{\pi/4} \tan^5 w\,dw$$

Step 2: Convert $\tan^5 w = \tan^3 w(\sec^2 w - 1) = \tan^3 w \sec^2 w - \tan^3 w$:

$$\int_0^{\pi/4} \tan^3 w \sec^2 w\,dw = \left[\frac{1}{4}\tan^4 w\right]_0^{\pi/4} = \frac{1}{4}(1^4 - 0) = \frac{1}{4}$$

Step 3: Evaluate $\int_0^{\pi/4} \tan^3 w\,dw = \int_0^{\pi/4} \tan w(\sec^2 w - 1)\,dw = \left[\frac{1}{2}\tan^2 w - \ln|\sec w|\right]_0^{\pi/4}$:

$$= \left(\frac{1}{2}(1) - \ln\sqrt{2}\right) - (0) = \frac{1}{2} - \frac{1}{2}\ln 2$$

Step 4: Combine results:

$$\int_0^{\pi/4} \tan^5 w\,dw = \frac{1}{4} - \left(\frac{1}{2} - \frac{1}{2}\ln 2\right) = -\frac{1}{4} + \frac{1}{2}\ln 2$$

Multiply by 2:

$$I = 2\left(-\frac{1}{4} + \frac{1}{2}\ln 2\right) = -\frac{1}{2} + \ln 2$$

$$\boxed{\int_0^{\pi/2} \tan^5 \frac{x}{2}\,dx = \ln 2 - \frac{1}{2}}$$

Problem 48: Evaluate $\displaystyle\int_0^{1/4} \sec \pi x \tan \pi x\,dx$

Method / Strategy: Direct integration using standard formula $\int \sec u \tan u\,du = \sec u + C$.

Step 1: Antiderivative is $\frac{1}{\pi}\sec \pi x$.

Step 2: Evaluate limits:

$$\left[\frac{1}{\pi}\sec \pi x\right]_0^{1/4} = \frac{1}{\pi}\left(\sec\frac{\pi}{4} - \sec 0\right) = \frac{1}{\pi}(\sqrt{2} - 1)$$

Step 3: Final Boxed Answer:

$$\boxed{\int_0^{1/4} \sec \pi x \tan \pi x\,dx = \frac{\sqrt{2} - 1}{\pi}}$$